

Implementing Load Balancing Using Pound



What we don't know or understand sometimes seems frightening. To set up a Web server takes a few minutes and a handful of commands—but if you're asked to implement load balancing on it, you may run short of ideas. In case you thought this requires a lot of time and expertise, this article will prove you wrong, because load balancing is quite simple.

Basically, load balancing is the process of setting up a number of Web servers for the same site, and distributing the Web traffic to these servers. Load balancing becomes a necessity for large sites because a single server cannot handle the flood of traffic alone. This technique is also used to reduce downtime and the potential loss of data in case of server failure.

So, after that brief introduction, let's get back to the point—how to implement load balancing!

Implementation

You'll need two systems for this exercise—one to run your Web servers, and one for running Pound, through which you will implement the balancing. This is when my admiration for VMWare increases, as I can create two virtual machines on my laptop, thus avoiding the need for a second system. You also need a package to install Pound, which you can download for the distribution you're using. I use RHEL6 for my Linux-related adventures, so I tried to find an *.rpm* package for Pound, but I ended up with a *.src.rpm* package. So the choice is yours—if you get an *.rpm* package (or any other package relevant to your distro) then install it as you usually do. But if you stick with my method of installing a *.src.rpm* package, then you'll learn an unorthodox way of installing packages in RHEL—which is an added advantage. Given below is the process of installing the Pound *.src.rpm* package:

- Install the *rpm-build* and *GCC* packages using the *yum -y install rpm-build gcc* command.
- Now issue *rpmbuild --rebuild /root/Desktop/pound-2.6.1.el5.src.rpm* (replacing the path with the location of the downloaded file).
- Traverse to the */root/rpmbuild/RPMS/i386/* directory. (I'm using a 32-bit machine; hence the directory is named *i386* under the *RPMS* directory. Yours may differ, depending on your machine's architecture.)

- Install the *.rpm* package with the familiar *rpm -ivh* command.

The first phase of the operation is over. Now it's time to relax a bit and switch to another system. Let's create two Web servers running a website, because a minimum of two Web servers are required to demonstrate the concept of load balancing. You don't need two machines to create two servers, as we'll use IP-based virtual hosting to run two Web servers on the same machine. The process is as follows; start with setting up a new network interface:

- Copy your interface configuration file to create another interface—for example, *cp /etc/sysconfig/network-scripts/ifcfg-eth3 /etc/sysconfig/network-scripts/ifcfg-eth3:0*. Here, my interface is *eth3*, so I've used *eth3* and *eth3:0* in my command. Check the appropriate interface on your system using the *ifconfig* command and use that one.
- Now edit the newly copied file and change the *DEVICE* and *IPADDR* settings. My *eth3* IP is 192.168.0.7 so I have changed to *DEVICE=eth3:0* and *IPADDR=192.168.0.8* in my *ifcfg-eth3:0* file.
- Bring down and bring up the interface using *ifdown eth3* and *ifup eth3* commands. Now, issuing the *ifconfig* command should show two interfaces *eth3* and *eth3:0* running with different IP addresses. In case you face a problem, issue *service NetworkManager stop*, and bring down and bring up the interface once again.
- Now, it's time to create the Web servers:
 - Create two directories: *mkdir /var/www/html/site{1,2}*
 - Create *index.html* files in each: in */var/www/html/site1/index.html* just enter "SITE1", and in */var/www/html/site2/index.html* "SITE2". Nothing fancy—just to check which site is serving the page.
 - Create entries in */etc/httpd/conf/httpd.conf*. You'll have to make the necessary changes as per your IP addresses, which I am sure you are smart enough to do.
- Here is my configuration: